

Lesson 2

IV and GMM, the Gauss–Markov theorem, and the geometry of least squares

1 The linear model and its assumptions

$$y = X\beta + u, \quad u \sim (0, \sigma^2 I_n),$$

with y : $n \times 1$, X : $n \times k$, β : $k \times 1$ and u : $n \times 1$. The same model is carried under two alternative sets of assumptions about the regressors and the errors, both keeping the error covariance spherical, plus a rank condition:

- X is fixed, $u \sim N(0, \sigma^2 I)$;
- X is stochastic, $u \sim (0, \sigma^2 I)$;
- $\text{rank}(X) = k$.

The brace the notes draw around the error statements is labelled *sphericity*: one common variance σ^2 , no correlation, which is what makes the algebra below simple. The third line is the rank condition that makes $X'X$ invertible, so that

$$b = (X'X)^{-1}X'y$$

is defined.

2 Orthogonality, and the need for instruments

Together with $\text{rank}(X) = k$, the two moment conditions that identify β are

$$\mathbb{E}(u) = 0, \quad \mathbb{E}(X'u) = 0 \quad (\text{the orthogonality condition}).$$

If the regressors are correlated with the error this fails. Then one looks for an instrument matrix

$$Z_{n \times p}, \quad p \geq k, \quad \mathbb{E}(Z'u) = 0,$$

that is, p variables that are uncorrelated with u but correlated with X (formally, $\text{rank } \mathbb{E}(Z'X) = k$). The moment condition $\mathbb{E}(Z'u) = 0$ is what replaces $\mathbb{E}(X'u) = 0$: it gives p equations for the k unknown elements of β .

3 The objective of an IV estimator

The idea is to choose β so that the sample moments $Z'u$ are as small as possible. As written on the page,

$$\min \|Z'u - \mathbb{E}(Z'u)\|, \quad \text{which with } \mathbb{E}(Z'u) = 0 \text{ is } \min \|Z'u\|.$$

Minimising a norm of a vector of moments is not the same as minimising the raw sum of squares, because the moments do not carry the same variance. Weighting by the inverse of their covariance (the metric of Lesson 1, $w'\Sigma^{-1}w$) gives

$$\min (Z'u)' [\text{Var}(Z'u)]^{-1} (Z'u).$$

Under sphericity $\text{Var}(Z'u) = Z'(\sigma^2 I_n)Z = \sigma^2 Z'Z$, so the inverse is $\sigma^{-2}(Z'Z)^{-1}$ and the objective becomes

$$\frac{1}{\sigma^2} u'Z (Z'Z)^{-1} Z'u \implies (y - X\beta)'Z (Z'Z)^{-1} Z'(y - X\beta),$$

the last step substituting $u = y - X\beta$. The factor σ^{-2} is a constant and drops out of the minimisation.

4 The generalized IV (two-stage least squares) estimator

Writing $P_z \equiv Z(Z'Z)^{-1}Z'$, minimise the objective over b :

$$\begin{aligned} & \min_b (y - xb)'Z (Z'Z)^{-1} Z'(y - xb) \\ &= \min_b (y - xb)'P_z(y - xb) \\ &= \min_b \left(y'P_z y - b'x'P_z y - y'P_z x b + b'x'P_z x b \right). \end{aligned}$$

The two middle terms are the same scalar, so together they are $-2b'x'P_z y$, and the first-order condition is

$$\begin{aligned} -2x'P_z y + 2x'P_z x b &= 0 \\ x'P_z y &= (x'P_z x) b, \end{aligned}$$

hence

$$b = (x'P_z x)^{-1} x'P_z y = (x'Z(Z'Z)^{-1}Z'x)^{-1} x'Z(Z'Z)^{-1}Z'y.$$

With more instruments than regressors, $p > k$ (the over-identified case), this is the *generalized IV* (two-stage least squares) estimator.

If instead there are exactly as many instruments as regressors, $p = k$, then Z is $n \times k$ and $Z'x$ is a square $k \times k$ matrix, invertible when the instruments are relevant, $\text{rank}(Z'x) = k$. Then $(x'Z(Z'Z)^{-1}Z'x)^{-1} = (Z'x)^{-1}(Z'Z)(x'Z)^{-1}$, and the same formula collapses to

$$b = (Z'x)^{-1}(Z'Z)(x'Z)^{-1} x'Z(Z'Z)^{-1}Z'y = (Z'x)^{-1}Z'y,$$

the *simple IV* estimator. With $p = k$ the choice of weights no longer matters: b sets the k sample moments $Z'(y - xb)$ exactly to zero.

5 Which estimator? A summary

Reading the assumptions as a table:

- $y = X\beta + u$ with $\mathbb{E}(X'u) = 0$ and $\text{Var}(u) = \sigma^2 I$: everything is spherical and the regressors are exogenous, so use least squares, $b = (x'x)^{-1}x'y$;
- $\mathbb{E}(X'u) \neq 0$: the orthogonality condition fails, so bring in instruments Z with $\mathbb{E}(Z'u) = 0$ and use IV, b_{iv} ; with $\text{Var}(u) = \sigma^2 I$ this is the estimator written out above;
- $\text{Var}(u) \neq \sigma^2 I_n$, i.e. $\text{Var}(u) = \sigma^2 \Omega_n$ with $\Omega \neq I_n$, while X is fixed and $\mathbb{E}(x'u) = 0$: the errors are non-spherical but exogenous, and the answer is *generalized least squares*;
- both failures at once, $\text{Var}(u) \neq \sigma^2 I_n$ and $\mathbb{E}(x'u) \neq 0$: generalized least squares is no longer consistent, and the answer is the *generalized method of moments* (GMM).

6 The OLS estimator in the normal linear model

Take the well-behaved case: $y = x\beta + u$ with $u \sim N(0, \sigma^2 I)$, X fixed and $\text{rank}(X) = k$, so that $x'x$ is invertible. The estimator is

$$\begin{aligned} b &= (x'x)^{-1}x'y \\ &= (x'x)^{-1}x'x\beta + (x'x)^{-1}x'u \\ &= \beta + (x'x)^{-1}x'u. \end{aligned}$$

With X fixed, b is β plus a linear function of u , so it inherits the distribution of u : it is unbiased,

$$\mathbb{E}(b) = \beta + (x'x)^{-1}x'\mathbb{E}(u) = \beta,$$

and its covariance is

$$\begin{aligned} \text{Var}(b) &= \mathbb{E}[(b - \beta)(b - \beta)'] = \mathbb{E}[(x'x)^{-1}x'uu'x(x'x)^{-1}] \\ &= \sigma^2(x'x)^{-1}, \end{aligned}$$

using $\mathbb{E}(uu') = \sigma^2 I_n$, so the sampling distribution is normal,

$$b \sim N(\beta, \sigma^2(x'x)^{-1}).$$

The estimator is the *BLUE* — best linear unbiased estimator — which is the content of the Gauss–Markov theorem, proved next. The theorem uses only $\mathbb{E}(u) = 0$ and $\text{Var}(u) = \sigma^2 I$; normality is not needed for it, only for the exact normal distribution of b .

7 Gauss–Markov: no other linear unbiased estimator beats OLS

Start from the model $y = X\beta + u$ and consider an arbitrary linear estimator

$$b^* = Cy, \quad C : k \times n, \quad y : n \times 1,$$

just as least squares $b = (X'X)^{-1}X'y$ is y premultiplied by the $k \times n$ matrix $(X'X)^{-1}X'$. Every such C can be written as the least-squares matrix plus a remainder,

$$C = [(X'X)^{-1}X' + A],$$

so that

$$b^* = Cy = [(X'X)^{-1}X' + A]y = (X'X)^{-1}X'y + Ay = b + Ay.$$

Substituting the model,

$$\begin{aligned} b^* &= (x'x)^{-1}x'(x\beta + u) + A(x\beta + u) \\ &= \beta + (x'x)^{-1}x'u + Ax\beta + Au. \end{aligned}$$

Taking expectations, $\mathbb{E}(u) = 0$:

$$\begin{aligned} \mathbb{E}(b^*) &= \beta + (x'x)^{-1}x'\mathbb{E}(u) + Ax\beta + A\mathbb{E}(u) \\ &= \beta + Ax\beta = \beta \quad \text{when } Ax = 0. \end{aligned}$$

So b^* is unbiased for every value of β exactly when A satisfies $Ax = 0$. That same condition — the one that restricts attention to the class of unbiased linear estimators — also removes the cross terms below. For the variance,

$$\begin{aligned}\text{Var}(b^*) &= \mathbb{E}[(b^* - \beta)(b^* - \beta)'] \\ &= \mathbb{E}\left[\left((x'x)^{-1}x'u + Au\right)\left(u'x(x'x)^{-1} + u'A'\right)\right] \\ &= \mathbb{E}\left[(x'x)^{-1}x'uu'x(x'x)^{-1} + Auu'x(x'x)^{-1} + (x'x)^{-1}x'uu'A' + Auu'A'\right] \\ &= \sigma^2(x'x)^{-1} + \sigma^2Ax(x'x)^{-1} + \sigma^2(x'x)^{-1}x'A' + \sigma^2AA'.\end{aligned}$$

With $Ax = 0$ the two cross terms vanish and

$$\begin{aligned}\text{Var}(b^*) &= \sigma^2(x'x)^{-1} + \sigma^2AA' \\ &= \text{Var}(b) + \sigma^2AA'.\end{aligned}$$

The extra term is σ^2AA' , which is positive semi-definite: for any $c \in \mathbb{R}^k$, $c'AA'c = \|A'c\|^2 \geq 0$. (It is not in general positive definite; it is nonzero whenever $A \neq 0$, and zero only when $A = 0$, i.e. $b^* = b$.) Adding it can only make the covariance larger, so no linear unbiased estimator has a smaller covariance than b : $\text{Var}(b^*) - \text{Var}(b)$ is positive semi-definite, i.e. for every c , $\text{Var}(c'b^*) \geq \text{Var}(c'b)$. Least squares is BLUE.

8 Geometric interpretation

In the model $y = X\beta + u$ the vector y is a point of $\mathbb{R}^n$, and the columns of X span a subspace

$$\dim L(x) = k$$

only k -dimensional. Write the least-squares decomposition as

$$y = Xb + e,$$

where the fitted value $Xb = X(X'X)^{-1}X'y$ lies in that subspace and $e = y - Xb$ is what is left over. The normal equations $X'(y - Xb) = 0$ say exactly that $X'e = 0$: the residual is orthogonal to every column of X . (The true $X\beta$ also lies in the subspace, but the error $u = y - X\beta$ is not orthogonal to it in general.)

Transcription notes

- The original opens with a title page reading only “Lesson 2”.
- The three error statements are grouped by a hand-drawn brace labelled *sphericity*; the arrangement is kept, the brace is not redrawn.
- The original mixes lower case (x) and upper case (X) for the regressor matrix, as in Lesson 1; the case is kept as written.
- $\mathbb{E}(Z'u) = 0$ is used on the same page as the condition $\min \|Z'u - \mathbb{E}(Z'u)\|$. Both are transcribed as written, though with $\mathbb{E}(Z'u) = 0$ the second term of the norm vanishes; the norm is then replaced by $(Z'u)'[\text{Var}(Z'u)]^{-1}Z'u$, which is what makes the weights appear.
- The constant σ^{-2} is carried into the last objective line here; the original drops it there, since it does not affect the minimiser.

orthogonal projection

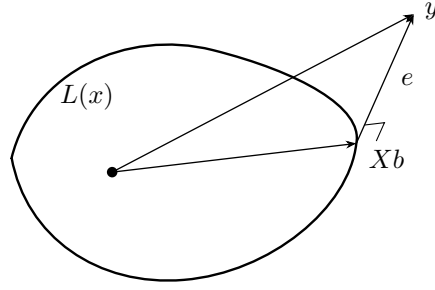


Figure 1: Xb is the orthogonal projection of y onto the k -dimensional subspace spanned by the columns of X , and the residual $e = y - Xb$ is orthogonal to that subspace. Minimising $\|y - X\beta\|^2$ over β is simply choosing the point of the subspace closest to y .

- **Corrected:** the first-order condition was written $-2b'x'P_z y + 2x'P_z x b = 0$; the derivative of $-2b'x'P_z y$ with respect to b is $-2x'P_z y$ (no b'), which is what the next line $x'P_z y = (x'P_z x)b$ uses.
- **Corrected:** in the just-identified collapse the middle line reads $(Z'x)^{-1}(Z'Z)x'(x'Z)^{-1}x'Z(Z'Z)^{-1}Z'y$ with a stray x' after $(Z'Z)$, and the line above has $x'Z'(Z'Z)^{-1}$ for $x'Z(Z'Z)^{-1}$; both are typeset without the slip.
- The page annotates the generalized-IV formula with “ $p > k$ ” and the simple-IV case with “ $K = p$, Z is $n \times K$ ”; the annotation is kept as “over-identified”/“just-identified”.
- $b \sim (\beta, \sigma^2(x'x)^{-1})$ is written on the page as the shorthand (mean, covariance); since u is normal the distribution is normal, and it is typeset $N(\beta, \sigma^2(x'x)^{-1})$.
- **Corrected:** the expectation of b^* was written $\mathbb{E}(b) + (x'x)^{-1}x'\mathbb{E}(u) + \dots$; $\mathbb{E}(b)$ already contains the term $(x'x)^{-1}x'\mathbb{E}(u)$, so the first term is β , not $\mathbb{E}(b)$.
- **Corrected:** “ $\sigma^2 AA'$ is positive definite when $A \neq 0$ ” is not true in general (A is $k \times n$ and AA' is positive definite only if A has full row rank k); the conclusion only needs positive semi-definiteness, which always holds.
- **Added:** the geometry page contains only $y = X\beta + u$, $y \in \mathbb{R}^n$ (written “ $y \in \mathbb{R}^n$ ”), $\dim \ell(x) = k$, the sketch and the words “orthogonal projection”. The decomposition $y = Xb + e$, the normal equations $X'e = 0$ and the remark on u are added explanation, as are the labels Xb , e and y in the figure (the sketch labels only the subspace).
- The just-identified case is written with K on the page; it is typeset $p = k$ to match the dimensions used everywhere else (k and K are the same number of regressors). The page states $\text{rank}(Z) = K$; what the formula needs is $\text{rank}(Z'x) = k$ (relevance), and the step showing the collapse to $(Z'x)^{-1}Z'y$ is added.
- On the last page the subspace is labelled $\ell(x)$ in the sketch (a cursive lower-case letter); it is the column space of X and is typeset $L(x)$.
- The sketch also has an arrow from the origin to the label $\ell(x)$ inside the subspace; it is read as a pointer to the label and not redrawn.
- The Gauss–Markov page writes “ $\text{Var}(b^*) = \sigma^2(x'x)^{-1} + \sigma^2 AA'$ where $A =$ ” and leaves the condition unfinished; it is $Ax = 0$, stated two lines earlier on the page.
- **Corrected:** the conclusion was written “for any, $\text{Var}(b^*) > \text{Var}(b)$ ”; the correct statement is that $\text{Var}(b^*) - \text{Var}(b)$ is positive semi-definite ($\geq$, with equality when $A = 0$).
- **Added:** the remarks that the moments are weighted by the inverse of their covariance (with the link to Lesson 1), that relevance means $\text{rank } \mathbb{E}(Z'X) = k$, that σ^{-2} drops out of the minimisation, that with $p = k$ the sample moments are set exactly to zero, that GLS is no longer consistent when $\mathbb{E}(x'u) \neq 0$, and that Gauss–Markov does not need normality, are explanation, not on the page.
- Pages of the original, for tracing back: page 2 the model and the assumptions, the IV moment condition

and the IV objective; page 3 the two-stage-least-squares and simple-IV formulas; page 4 the summary of assumptions and estimators and the start of $b = \beta + (x'x)^{-1}x'u$; page 5 $\mathbb{E}(b)$, $\text{Var}(b)$ and the start of b^* ; page 6 the Gauss–Markov proof; page 7 the geometric interpretation.